MOBL-99: Best Practice Summary

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Overview

This notebook consolidates every actionable best practice from the MOBL series (notebooks 01 through 12) into a single, definitive reference. Each best practice specifies the exact setting or value to use, its priority level, and which category it belongs to. Use this as a checklist when instrumenting, configuring, and operating Dynatrace native mobile monitoring.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled

Requirement	Details
Permissions	`storage:user.events:read`, `storage:user.sessions:read`, `storage:smartscape:read`; `storage:bizevents:read` for custom business events
Mobile App	At least one mobile application with Dynatrace SDK integrated
Prior Knowledge	Familiarity with MOBL-01 through MOBL-12 recommended

1. SDK Setup & Configuration

Best practices for installing and configuring the Dynatrace mobile SDK across iOS, Android, and cross-platform frameworks.

#	Best Practice	Recommended Setting/Value	Priority	Source
0	Turn on the New Real User Monitoring Experience	Environment: Settings > Collect and capture > Real User Monitoring > Enablement and cost control > Mobile > **Enable RUM**; frontend: Settings > Enablement and cost control > **New Real User Monitoring Experience**. Without it the `user.events` / `user.sessions` queries return nothing	**Critical**	MOBL-02, MOBL-03
1	Enable auto-start for production apps	iOS: `DTXAutoStart = true` in Info.plist; Android: `autoStart { applicationId(...); beaconUrl(...) }` inside a named configuration in the top-level Gradle build file	**Critical**	MOBL-02, MOBL-03
2	Keep crash reporting on	iOS: `DTXCrashReportingEnabled = true`; Android: on by default -- do not set `crashReporting(false)`	**Critical**	MOBL-02, MOBL-03
3	Use Swift Package Manager over CocoaPods for iOS	Add via Xcode: File > Add Package Dependencies	**Recommended**	MOBL-02
4	Control the Gradle plugin version deliberately	Docs recommend `version "8.+"` in the top-level `plugins {}` block (minor updates automatic, majors by hand); pin an exact `8.x.y` if you need reproducible builds	**Critical**	MOBL-03
5	Fill in both platform sections of a cross-platform config	Use the Android and iOS values the instrumentation wizard gives you for each section	**Critical**	MOBL-04

#	Best Practice	Recommended Setting/Value	Priority	Source
6	Use separate Application IDs for prod vs staging	Android: variant-specific configurations matched with `variantFilter`; iOS: different Info.plist per scheme	**Recommended**	MOBL-03
7	Never commit Application IDs or Beacon URLs to public repos	Store in environment variables or `local.properties` excluded from version control	**Critical**	MOBL-03
8	Enable hybrid monitoring for WebView apps	iOS: `DTXHybridApplication = true`; Android: `hybridWebView { enabled(true) }`	**Recommended**	MOBL-02, MOBL-03
9	For Flutter: use the downloaded `dynatrace.config.yaml` at project root	Apply it with `dart run dynatrace_flutter_plugin`, and start with `Dynatrace().start(MyApp())`	**Critical**	MOBL-04
10	For React Native: re-run `npx instrumentDynatrace` after config changes	Changing `dynatrace.config.js` does not auto-apply; the instrument command must re-run	**Critical**	MOBL-04
11	Verify both platforms report for cross-platform apps	Group `user.sessions` by `os.name` and confirm both Android and iOS sessions appear; list frontends with `smartscapeNodes "FRONTEND" | filter frontend.type == "mobile"`. Classic equivalent: `fetch dt.entity.mobile_application` -- **not** `dt.entity.device_application`, which does not exist and returns zero rows	**Critical**	MOBL-04

2. Crash Reporting & Symbolication

Best practices for ensuring crash data is complete, readable, and actionable.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Upload dSYM files for every iOS release build	Preprocess with `DTXDssClient -decode`, then upload (`DTXDssClient -upload`, Mobile Symbolication API, Fastlane plugin, or web UI)	**Critical**	MOBL-06

#	Best Practice	Recommended Setting/Value	Priority	Source
2	Upload ProGuard/R8 mapping files for every Android release build	Not automatic in the Gradle build: use the DSSClient, the Mobile Symbolication API (`DssFileManagement` token), the Fastlane plugin, or the web UI — from CI	**Critical**	MOBL-03, MOBL-06
3	Upload React Native source maps	Settings > Web and mobile monitoring > Source maps and symbol files > React Native > Upload files (bundle file, name, version)	**Critical**	MOBL-04
4	Verify symbolication status after upload	Settings > Web and mobile monitoring > Source maps and symbol files; pin files you must keep (1 GiB quota); stack traces must show method names, not hex addresses	**Critical**	MOBL-06
5	For Bitcode iOS builds: download dSYMs from App Store Connect	Local dSYMs do not match Apple-recompiled binaries; download from Xcode Organizer	**Critical**	MOBL-06
6	Leave OneAgent's R8 keep rules intact	They ship with the agent and R8 applies them; configure a third-party obfuscator to honor them, and do not filter them out (`ignoreFrom`)	**Critical**	MOBL-03
7	Report handled exceptions via SDK API	iOS: `DTXAction.reportError(withName:error:)` ; Android: `Dynatrace.reportError(name, exception)`	**Recommended**	MOBL-06
8	Use descriptive error names for reported errors	`"Payment Failed"` not `"Error"` -- enables meaningful DQL grouping	**Recommended**	MOBL-06
9	Report errors at catch boundaries, not in utility methods	Attach full context (user action, screen name) where the error is handled	**Recommended**	MOBL-06
10	Do not report transient/expected conditions as errors	Network retries, expected timeouts, and recoverable conditions create noise	**Recommended**	MOBL-12
11	Set a crash-free target from your baseline	Community starting bands: > 99.5% healthy, 98-99.5% review, < 98% act; Google Play's user-perceived crash threshold is 1.09% (per user, not per session)	**Critical**	MOBL-10, MOBL-11

3. User Action Instrumentation

Best practices for tracking user interactions accurately and meaningfully.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Instrument SwiftUI with the SwiftUI instrumentor	`brew install DTSwiftInstrumentor`, then `DTSwiftInstrumentor install`; add it to CI builds	**Critical**	MOBL-02, MOBL-05
2	Rely on Jetpack Compose auto-instrumentation	On by default from Android Gradle plugin 8.271; add `enterAction` / `leaveAction` only for business-level flows	**Critical**	MOBL-03, MOBL-05
3	Always call `leaveAction()` in a finally block	Forgetting to close an action inflates duration metrics and orphans child events	**Critical**	MOBL-03, MOBL-05
4	Use descriptive, stable action names	`"Cart: Add Item"` not `"btn_click"` or `"button_42"`	**Critical**	MOBL-05
5	Never put dynamic values in action names	`"View Product Details"` not `"View Product #48291"` -- dynamic values cause cardinality explosion	**Critical**	MOBL-05
6	Never put user IDs or timestamps in action names	Creates infinite unique names, defeats grouping, and is a privacy risk	**Critical**	MOBL-05
7	Use feature-area prefixes in naming convention	`"Cart: Add Item"`, `"Search: Apply Filter"`, `"Checkout: Submit Order"`	**Recommended**	MOBL-05
8	Use consistent casing across platforms	`"Search Products"` on both iOS and Android, not mixed case variants	**Recommended**	MOBL-05
9	Exclude SwiftUI controls that add noise	The SwiftUI instrumentor supports global and local exclusion of controls	**Recommended**	MOBL-02
10	Configure user action naming rules in Dynatrace UI	Mobile app settings > Naming rules (naming or extraction rules on the generated name); test against recent data before deploying	**Recommended**	MOBL-05
11	Keep action hierarchies shallow	Parent + one level of child actions (the Flutter plugin allows no deeper)	**Recommended**	MOBL-05, MOBL-12

4. Network Request Monitoring

Best practices for HTTP monitoring, backend correlation, and network performance optimization.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Instrument backend services with OneAgent, OpenTelemetry, or both	Required for `x-dynatrace` / W3C trace-context correlation (W3C from OneAgent for Mobile 8.333); enables end-to-end distributed tracing from device to database	**Critical**	MOBL-07
2	Use manual web request tagging for non-standard transports	Add the header named by `Dynatrace.getRequestTagHeader()` (`x-dynatrace`) with a tag from `action?.getTagFor(url)` (iOS) or `action.getRequestTag()` / `Dynatrace.getRequestTag()` (Android) for WebSocket, gRPC, or custom HTTP clients	**Recommended**	MOBL-12
3	Enable gzip/brotli compression on API responses	Substantially smaller text payloads	**Recommended**	MOBL-07
4	Implement response pagination	Avoid loading entire datasets in a single request	**Recommended**	MOBL-07
5	Use HTTP/2 or HTTP/3 for multiplexing	Reduces connection overhead and round trips	**Recommended**	MOBL-07
6	Implement client-side caching with ETags and Cache-Control	Prevents redundant network requests for unchanged resources	**Recommended**	MOBL-07
7	Serve static assets from a CDN	Reduces DNS, TCP, and TLS times for geographically distributed users	**Recommended**	MOBL-07
8	Adapt app behavior to connection type	Reduce image quality, defer non-critical requests, enable offline queueing on slow connections	**Optional**	MOBL-07
9	When the backend span dominates a request's `duration`, investigate the backend	Use frontend-to-backend correlation to trace into server-side services and database calls; mobile request events carry no DNS/TCP/TLS breakdown	**Recommended**	MOBL-07
10	Exclude analytics/CDN URLs from SDK monitoring	iOS `DTXURLFilters` in Info.plist; Android `urlFilters` in the plugin's `webRequests` block (8.339+)	**Recommended**	MOBL-12

5. Session Replay & Privacy

Best practices for configuring mobile session replay with appropriate privacy controls.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Enable session replay on crash	App settings > General > Enablement and cost control > **Enable Session Replay on crashes** (no client-side key)	**Critical**	MOBL-08
2	Choose the masking level deliberately	Safest is the default; to use Safe, set `MaskingConfiguration(maskingLevelType: .safe)` (iOS) or `MaskingConfiguration.Safe()` (Android) in code	**Critical**	MOBL-08
3	Set the production Full Session Replay percentage to 5-10%	App settings > General > Enablement and cost control; there is no SDK sample rate	**Critical**	MOBL-08
4	Set staging/QA sample rate to 100%	Full coverage for pre-release testing and bug verification	**Recommended**	MOBL-08
5	Use Safest masking for financial/healthcare apps	Replays show layout and navigation but no readable content	**Critical**	MOBL-08
6	Only use Custom masking when you need to unmask specific elements	Start with Safest or Safe, move to Custom only for targeted debugging; mask views via `MaskingConfiguration` (`addMaskedView` / `addMaskedIds`), `Modifier.dtMask()` in Compose, or the `data-dtrum-mask` tag	**Recommended**	MOBL-08
7	Temporarily increase sample rate during incidents	Raise to 50-100% when actively investigating a reported issue; lower again after resolution	**Recommended**	MOBL-08
8	Monitor Session Replay consumption	Track it in the license overview (DEM units on classic licensing; your DPS rate card otherwise)	**Recommended**	MOBL-08
9	Limit what session replay captures (masking, sampling)	Replay retention is fixed at 35 days in a built-in bucket that cannot currently be modified	**Optional**	MOBL-09

6. Session Properties & User Tagging

Best practices for enriching sessions with business context and identifying users.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Tag users with opaque IDs, never PII	`Dynatrace.identifyUser("usr_a1b2c3d4")` -- never email addresses, phone numbers, or full names	**Critical**	MOBL-09
2	Clear user tag on logout	iOS: `Dynatrace.identifyUser(nil)`; Android: `Dynatrace.identifyUser(null)`	**Critical**	MOBL-09
3	Report session-property values early in the session	Call `action.reportValue()` on an early user action (values must be part of a user action)	**Recommended**	MOBL-09
4	Keep the property set small and deliberate	Every property is data you send, store and must justify under data minimization	**Recommended**	MOBL-09
5	Use enum-like string values for properties	`"tier" = "free"` not `"tier" = "Free Trial Account"` -- enables clean aggregation	**Recommended**	MOBL-09
6	Use consistent property key names across iOS and Android	Same keys on both platforms so a single DQL query covers both	**Recommended**	MOBL-09
7	Never store PII in session property values	A property is stored as sent — treat it like any field you would have to delete on request	**Critical**	MOBL-09
8	Report A/B test variants as session properties	`action.reportValue("ab_test_checkout_v2", "variant_b")` on an open action -- enables correlation with performance/crash data	**Recommended**	MOBL-12

7. Privacy & Compliance

Best practices for GDPR, CCPA, and general data privacy compliance.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Implement opt-in mode for GDPR compliance	`DTXUserOptIn = true` (iOS) or `userOptIn(true)` (Android); call `applyUserPrivacyOptions` after user consent	**Critical**	MOBL-09
2	Collect nothing until the user chooses	In opt-in mode OneAgent sends only a single "Loading " event (device metadata) until `applyUserPrivacyOptions` is called -- turn off auto-start and start manually after consent if nothing may be sent; `PERFORMANCE` is the anonymous level a user can choose, `USER_BEHAVIOR` the full one	**Critical**	MOBL-09
3	Rely on OneAgent to persist the applied preferences	OneAgent persists privacy preferences and re-applies them on restart; the app only needs to ask when no choice exists yet	**Critical**	MOBL-09
4	Provide a way to withdraw consent in app settings	Set data collection level to `Off` when user revokes consent	**Critical**	MOBL-09
5	Handle right-to-erasure requests with the documented deletion routes	Sensitive Data Center deletion request (UI) or Grail record deletion (API, covers `user.events`, `user.sessions`, `user.replays`)	**Critical**	MOBL-09
6	Minimize retained personal data	RUM data sits in built-in 35-day buckets that cannot currently be modified; set short retention only on data you route yourself (business events) and minimize what the SDK captures	**Recommended**	MOBL-09
7	Separate crash reporting opt-in from general monitoring	Crash reporting has its own `crashReportingOptedIn` flag independent of data collection level	**Recommended**	MOBL-09
8	Display clear, plain-language consent dialog	Explain what data is collected, why, and how; provide granular options for Performance vs User Behavior levels	**Critical**	MOBL-09
9	Do not plan per-region RUM buckets	Custom RUM buckets cannot currently be created, so RUM data cannot be split into EU and non-EU buckets with OpenPipeline	**Optional**	MOBL-09

8. Dashboards & Alerting

Best practices for operationalizing mobile monitoring with dashboards and alerts.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Include a crash-free rate single-value tile with color thresholds	Community starting bands: green > 99.5%, yellow 98-99.5%, red < 98% — tune to your baseline	**Critical**	MOBL-11
2	Organize dashboard: health at top, trends in middle, drill-down at bottom	Top: single-value tiles (crash rate, sessions); Middle: timeseries (trends); Bottom: tables (top crashes, slowest actions)	**Recommended**	MOBL-11
3	Add a dashboard variable for `frontend.name`	Lets stakeholders filter to their specific app	**Recommended**	MOBL-11
4	Set default time range to 24 hours with presets for 1h, 6h, 24h, 7d	Covers most operational use cases without excessive data scanning	**Recommended**	MOBL-11
5	Alert on high crash rate	Davis anomaly detector on a DQL crash query (`characteristics.has_crash`), static threshold such as > 10 crashes per hour	**Critical**	MOBL-11
6	Alert on session volume drop	< 50% of rolling 7-day baseline, severity Warning	**Recommended**	MOBL-11
7	Alert on slow app launch	Average app start > 5 seconds over a 15-minute window, severity Warning	**Recommended**	MOBL-11
8	Alert on high HTTP error rate	> 5% of mobile requests returning 5xx, severity Critical	**Critical**	MOBL-11
9	Use Dynatrace Intelligence for adaptive baselines on performance metrics	Dynatrace Intelligence automatically learns patterns; add static thresholds only for hard SLA limits	**Recommended**	MOBL-11
10	Tag mobile app entities with `app-type: mobile`	Enables the Problem trigger's affected-entity tag filter for mobile-specific problems (or use the custom filter `matchesValue(affected_entity_ids, "MOBILE_APPLICATION-*")`)	**Recommended**	MOBL-11
11	Create audience-specific dashboards	Developers: 5-min refresh, crash detail; QA: 15-min, crash-free rate by version; Executives: daily KPI snapshot	**Recommended**	MOBL-11
12	Break down crash rate by `app.short_version` after every release	Identifies if a specific release introduced a regression	**Critical**	MOBL-10, MOBL-11

9. SDK Performance Optimization

Best practices for minimizing SDK overhead on battery, network, and CPU.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Exclude third-party analytics and CDN URLs from monitoring	iOS `DTXURLFilters` (Info.plist array, wildcards); Android `urlFilters` in `webRequests` (8.339+)	**Recommended**	MOBL-12
2	Use cost and traffic control for high-traffic production apps	App settings > General > Enablement and cost control; a lower monitored-session share reduces DEM consumption	**Recommended**	MOBL-08, MOBL-12
3	Start with full monitoring, reduce only if needed	Do not prematurely optimize; measure SDK overhead first	**Recommended**	MOBL-12
4	Set an app-launch target from your baseline	Community rule-of-thumb bands: < 1s excellent, 1-2s acceptable, 2-5s needs work, > 5s investigate; re-baseline after OneAgent for Mobile 8.347 caps inflated cold starts	**Critical**	MOBL-10, MOBL-11

10. Advanced Instrumentation

Best practices for custom business events, A/B testing, multi-app strategies, and advanced SDK usage.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Use reverse-domain naming for custom business event types	`"com.myapp.purchase"` not `"purchase"` -- avoids collisions across apps and services	**Recommended**	MOBL-12
2	Send business events for conversion-critical interactions	Purchases, sign-ups, feature usage via `sendBizEvent()` with structured attributes	**Recommended**	MOBL-12
3	Report feature flag evaluations as business events	`event.type = "com.myapp.feature_flag"` with `feature.name` and `feature.variant` attributes	**Optional**	MOBL-12
4	Use nested actions for multi-step flows	Parent action wraps the full flow; child actions measure individual steps (validate, submit, confirm)	**Optional**	MOBL-12

#	Best Practice	Recommended Setting/Value	Priority	Source
5	For long-running actions, manage lifecycle explicitly	Call `leaveAction()` in completion handler or `finally` block; do not rely on auto-close timeout	**Critical**	MOBL-12
6	Use consistent attribute names across iOS and Android business events	Same keys on both platforms enable unified DQL queries	**Recommended**	MOBL-12
7	Never include PII in business event attributes	No email addresses, phone numbers, or personal data in custom event payloads	**Critical**	MOBL-12
8	Use consistent naming prefixes for multi-app organizations	`"MyBrand Customer"`, `"MyBrand Driver"`, `"MyBrand Admin"` for easy DQL filtering	**Recommended**	MOBL-12
9	Budget DEM units per app based on criticality	100% monitoring for critical apps; use cost and traffic control for high-traffic secondary apps	**Recommended**	MOBL-12
10	Attach `reportValue()` properties to actions for business context	`action?.reportValue(withName: "total_price", doubleValue: 49.99)` -- convert to action/session properties in the app settings	**Recommended**	MOBL-12

11. DQL Query Patterns

Mandatory patterns and filters for querying mobile data in Grail.

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Query mobile RUM from `user.events` / `user.sessions`	`filter dt.rum.application.type == "mobile"` separates mobile from web; `characteristics.has_*` selects the event type. Mobile RUM is **not** in `bizevents`	**Critical**	MOBL-10
2	Count sessions on `user.sessions`	One record per session, so `count()` is a session count there; on `user.events` use `countDistinct(dt.rum.session.id)`	**Critical**	MOBL-10
3	Compute crash rate at session grain	`fetch user.sessions` then `countIf(error.has_crash == true)` over `count()`	**Recommended**	MOBL-10

#	Best Practice	Recommended Setting/Value	Priority	Source
4	Add `isNotNull()` filters before grouping on optional fields	Fields like `app.short_version`, `device.model.identifier`, `device.manufacturer` may be null; filter first to avoid noise	**Recommended**	MOBL-10
5	Query mobile app entities for inventory	Preferred: `smartscapeNodes "FRONTEND" | filter frontend.type == "mobile"`. Classic fallback (still functional): `fetch dt.entity.mobile_application`. No time range needed; returns current state. `dt.entity.device_application` does not exist -- it returns zero rows, not an error	**Recommended**	MOBL-10
6	Filter crashes with `characteristics.has_crash`	Reported errors are `characteristics.has_error` with `characteristics.is_api_reported` -- distinguish between the two	**Critical**	MOBL-06, MOBL-10
7	Always specify explicit time ranges	`fetch user.events, from:-1h` or `from:-24h` -- never rely on the default 2-hour window	**Critical**	MOBL-10
8	Segment by `app.short_version` for release monitoring	Enables crash rate and performance comparison across releases	**Recommended**	MOBL-10

Priority Summary

Priority	Count	Description
Critical	47	Must implement -- failure to do so causes data loss, security risk, broken monitoring, or compliance violation
Recommended	52	Should implement -- significantly improves data quality, operational efficiency, or user experience
Optional	5	Nice to have -- advanced capabilities for mature mobile monitoring programs

Implementation Order

1. **First**: All Critical items in SDK Setup, Crash Reporting, and Privacy sections
2. **Second**: Critical items in User Action Instrumentation and Dashboards/Alerting

3. **Third**: All Recommended items
4. **Last**: Optional items for advanced use cases

References

- Dynatrace Mobile Monitoring Documentation
- iOS SDK Integration Guide
- Instrument Android apps (DT docs)
- Session Replay for Mobile
- Mobile SDK Privacy Settings

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.